

Galactic kinematics of the Solar neighborhood

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Simulation of the dynamical behavior of a synthetic population of white dwarfs belonging to thin disk, thick disk, and halo making use of the software galpy. We compared, analyzed, and characterized the three different types of orbits using important parameters such as apoapsis, z maximum, and guiding-center radius, among others.

I. INTRODUCTION

The Milky Way has remained a paramount subject of study in astronomy. Once believed to encapsulate the entirety of what we understood as the universe, presently, we recognize it as merely one among many spiral galaxies residing within a localized assemblage known as the Local Group. This, in turn, is but one of the numerous congregations that comprise the grandeur of the Virgo Cluster. Yet, on an even vaster scale, lies the Laniakea supercluster, the most immense structure to which we belong, where the universe begins to exhibit homogeneity and isotropy.

Aside from the comparatively small scale, the study of the Milky Way is fundamental for our comprehension of the entire universe. Within our galaxy, we discern three principal types of galaxies: elliptical nebulae, spirals, and lenticular formations. Our spiral galaxy boasts a bulge, a disk, and a stellar halo. The disk could be subdivided into thin disk and a thick disk with different dynamical properties [8]. These last two parts of the galaxy along with the halo will be studied in detail in this project.

Understanding the orbits of celestial bodies within our galaxy affords us the means to study its evolution, encompassing both chemical and gravitational properties. Chemical traits, such as metallicity, provide insight into a star's birthplace, while dynamic properties, including mass distribution, stand as fundamental pillars in comprehending elusive concepts like dark matter or dark energy.

II. MATHEMATICAL FOUNDATIONS

The next description is entirely based on the workshop for this project, see [7].

First, the number of elements or white dwarfs simulated is 15000. This number as we expect has an important weight in the code-runtime.

Second, our stars belong to the solar neighborhood ($<100\text{pc}$), so the center of the random spatial uniform distribution is $X = 8\text{ kpc}$, $Y = 0\text{ kpc}$, $Z = 0.025\text{ kpc}$.

Third, space velocities are different for each type of population and direction, we used velocities found in Table 1 in the guide [7].

Fourth, circular velocity and solar motion. As we said Milky Way is a spiral galaxy. Like every galaxy, it rotates. Due to the use of the galactocentric coordinates, we include a circular velocity of $V_o = 220\text{ km/s}$ in all our objects because we assumed an average velocity for objects about $R_o = 8\text{ kpc}$ distant from the center of the galaxy. In addition, we included the solar motion as described in the guide ($U_\odot = -11.1\text{ km/s}$, $V_\odot = 7.25\text{ km/s}$, $W_\odot = 12.24\text{ km/s}$) [7].

Finally, to integrate orbits by using the potential MWPotential2014 [1] from galpy we also have to define some physical parameters such as the time-lapse and time step, chosen in 2 gigayears and 1 Myr, which is enough to look into the behavior of the orbits but is not computationally problematic ??.

III. ANALYSIS AND RESULTS

A. Toomre diagram

The computational work of this project consisted of modifying the code given in class to allow the integration of several orbits at the same time. Moreover, we introduced a uniform spatial spherical distribution for the white dwarfs around the solar neighborhood using the parameters mentioned previously (radius and center). Furthermore, new intriguing relationships have been plotted and will be presented. The edited code can be found here¹

At an initial glance at the diagram 1, a seeming contradiction arises: the mean velocity appears to be less than zero, conflicting with the kinematic definition of

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¹ follow the path Computational Astro /Programs /Simulated orbits copia ipynb here.

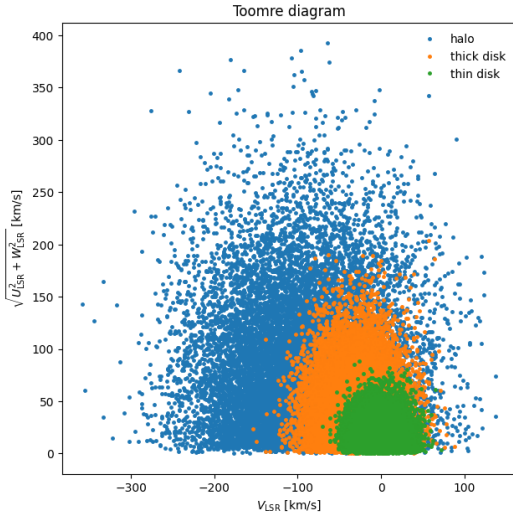


Figure 1. The Toomre diagram of the 15000 stars of each kind. In the x-axis is plotted the circular or angular velocity of the stars, meanwhile in the y-axis is shown the euclidean norm of the other two velocities(radial and vertical). In green are colored the white dwarfs in the thin disk, orange for thick, and blue for the halo white dwarfs

the local standard of rest, which denotes the average motion of stars nearby to the sun. However, this apparent paradox is resolved when considering that approximately 85% of the nearby stars in the sun's vicinity belong to the thin disk category [4], 25% to the thick disk and only 0.1 to the halo[3]. Due to thin disk stars exhibit velocities represented by a normal distribution, centered at -1.45 km/s, close to the mean velocity selected for our kinematic definition[7], thereby reconciling the observed discrepancy.

An interesting aspect lies in examining the vertical and radial norms, offering insights into deviations from a perfect circular orbit. The local standard frame of rest is instantaneously centered on the sun but instead of following the sun's no zero eccentric orbit ($e_\odot < 0.1$), follows a perfect circular orbit[8]. Also, the vertical velocity of the stars is a measure of the inclination of the orbit. Taking this into account we can state that stars in the thin disk have low eccentricities and low inclinations, as we would expect because they are located in a thin region close to the plane of the galaxy of about 300-400 parsecs[4].

As theory claims sun ($V_\odot = 7.25 \text{ km/s}$) behaves as a star located in the thin disk. However is a little bit faster than the average of the stars in the thin disk and consequently than the average of stars in the diagram.

For stars situated in the thick disk, their distribution centers around -30. While it might suggest a counterclockwise rotation within the galaxy, it's essential to consider our position in the local standard of rest, in

motion at an intrinsic velocity of 220 kilometers per second. This perspective indicates that the objects under analysis merely exhibit an average intrinsic velocity that is 30 km/s slower. Additionally, the inclination concerning the galactic plane is more pronounced, resulting in an expanded region characterized by the z max value analyzed in III C.

On average, halo stars exhibit even lower velocities compared to thick-disk stars. However, their dispersion is considerably wide, resulting in some of these stars being the fastest stars in the diagram. Remarkably, as we explore in subsequent sections, certain stars possess significantly large and negative relative velocities, enabling them to move in a counterclockwise direction against the galaxy's rotational motion.

B. Angular momentum and eccentricity

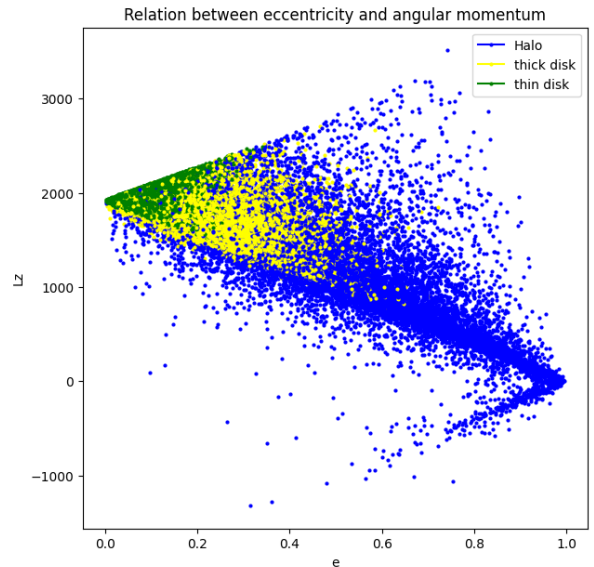


Figure 2. Relation between the angular momentum of the white dwarfs and the eccentricity in which blue are the halo stars, yellow for thick disk stars, and green for thin disk stars

In depiction 2 the majority of the points are located in a triangular region with a clear vertex in zero eccentricity except for some rebel halo stars, specifically the density increase at the edges of this imaginary triangle.

As we would expect for stars in a close orbit(elliptical) no white dwarf exceeds an eccentricity of 1 because conic sections with eccentricities above 1 are hyperbole. Indeed, stars in the thin and thick disks do not even reach $e = 1/2$.

As mentioned previously in section III A some halo stars have a very large velocity in the opposite direction to the sun. In diagram 2 we check out that some

stars appear with negative angular momentum, 0.027% is the percentage of halo stars that follow retrograde orbits. Stars in the thin and thick disks do not show this behavior. However, this number is big enough to concern us about the reason for this apparently unnatural motion.

In our solar system most objects orbit in the same direction, the same direction of the suns revolves. This happens because the solar system was born from the same cloud of dust with a specific angular momentum that has to be conserved. This same formation process is followed by larger astronomical structures. Therefore, a star that rotates counter wise the normal galaxy motion was born differently than the stars in the thin and thick disk. One hypothesis is that those stars come from the accretion of a single massive dwarf galaxy, the Gaia Sausage [6].

C. Spatial distribution

First, we analyze a low number of objects with a large integration time 3, then we proceed with the total sample of stars 4.

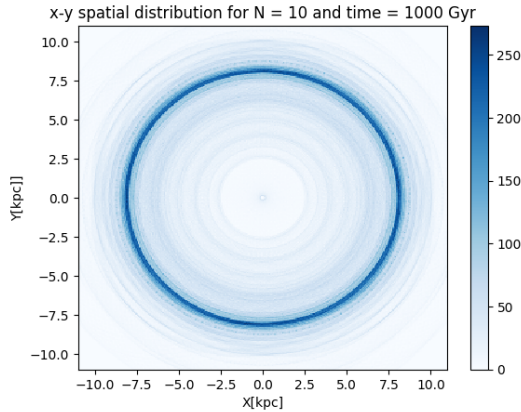


Figure 3. Two-dimensional histogram of the orbits of a short population of 1 stars, of each kind, over 2 gigayears from bird's-eye view x-y

To characterize these regions we should analyze the pericenter and apocenter of distinct orbits I

	Apoapsis[kpc]	Periapsis[kpc]
Halo	10.13	3.40
Thin disk	8.44	7.15
Thick disk	9.10	6.06

Table I. Average value of the apoapsis and periapsis of the star populations

As expected by the low eccentricity thin disk stars follow an almost circular path. With pericenter and

apocenter very close to each other, checking out that these stars are represented by the solid dark blue ring at 8 kpc. This region is enclosed by the thick disk which has a higher average eccentricity and consequently a larger dispersion, being one parsec larger than the thin region 6 to 9 parsecs.

Halo stars have more eccentric orbits 2, consequently farther apoapsis and closer periapsis to the center of the galaxy which is represented in the diagram as the region beyond the solid line of 9 kpcs and the rings before 6 kpcs. This is the larger and more homogeneous region, being possible even to observe dim circles close to the galactic center.

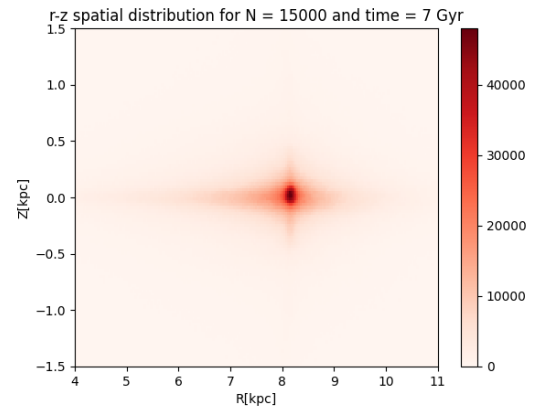


Figure 4. Two-dimensional histogram of the orbits of the entire population of stars 15000 of each kind of star over 2 gigayears from side-on z-r

To characterize these regions a prior analysis over the z maximum and orbit inclination of distinct orbits must be done I

	Z max[kpc]	Verticle velocity[km/s]
Halo	1.73	-8.54
Thin disk	0.22	-3.46
Thick disk	0.53	0.25

Table II. Average value of the z maximum and vertical velocity for each kind of star

The Toomre diagram 1 showed thin disk stars have slow vertical velocities, consequently low sharp vertical angles and not too high z maximumII. For stars in the thick disk these parameters are higher which results in a bigger region in which these stars are located, roughly 4.8 kpc [5].

In the image, we can observe the galactic plane in which is located the majority of the stars, especially the ones that combine the thin and thick disks. Remarkably, a much denser region remains close to the sun's radial coordinate.

As usual halo stars are exotic, with the highest z maximum values we can ascribe them to the complex orbits above the galactic plane, even almost vertical orbits. Consequently, the dim vertical line in 8kpc.

Actually due to the steep nature, orbits some of them go through the galactic disk causing the intersection with sun orbits as the class M1 red subdwarf Kapteyn, which is the closest halo star to the solar system [2]

D. Guiding center

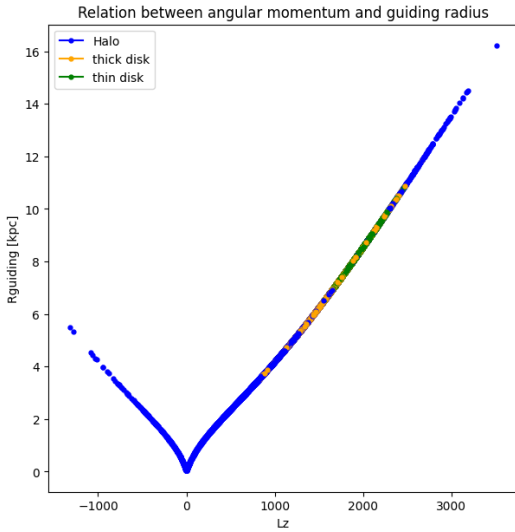


Figure 5. Plot of the guiding radius center as a function of the angular momentum for a sample of 15000 of each kind of star and integration over 7 Giga years

In depiction 5 we can observe a perfect linear correlation between the guiding center radius and the angular

momentum because the guiding center radius is defined as the radius of a circular orbit with the same angular momentum. Surprisingly, the diagram shows that some orbits of the halo region have an angular momentum small enough to achieve a zero guiding radius. A possible explanation is that these stars may have originated in the vicinity of the galactic center, and their orbit may have been altered by a supernovae.

IV. CONCLUSIONS

In conclusion, through the integration and spatial representation of the orbits of a substantial sample of 15000 stars over 7 Gyr, it was possible to characterize the kinematic behavior and properties of these three kinds of stars.

Making use of the spatial distribution diagrams and some properties as the z maximum, periapsis, and apoapsis we were enabled to describe the spatial domains of each kind of star. In summary, the thin disk receives its name due to the low vertical motion and small eccentricities that lead to circular-like orbits. The thick disk is a wider region in vertical and radial dimensions. The galactic halo is a much bigger zone the eccentricity of their orbits allows stars to reach the galactic center or far beyond the galactic disk.

However, we were unable to determine the theoretically spherical spatial distribution of the galactic halo, as the sample is confined to the solar vicinity in time zero.

In addition, through analyzing the negative angular momentum and the velocity distribution of some halo stars we conclude that they follow retrograde orbits. Moreover, we studied the guiding radius and angular momentum relation.

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